AI-driven Automation for Data Systems

PhD Thesis Overview · Saeed Fathollahzadeh · Concordia University · June 2026

Common Approach: Zero-Shot In-Context Learning (ICL) · Metadata-Aware LLM Prompting · Automated Validation & Verification

Chapter 1: CatDB

LLM-guided ML Pipeline Generation

Problem: Generating effective data-centric ML pipelines for unseen datasets is labor-intensive; LLMs hallucinate without dataset context.

Approach:

- Data catalog profiling & metadata refinement
- Dataset-specific zero-shot prompt construction
- Automated error handling & pipeline repair

Result: Up to $\times 10$ faster than AutoML; matches GPT-4 accuracy on 20 diverse datasets.

Publications

PVLDB 2025 · SIGMOD 2025

Chapter 2: ReSequel

LLM-assisted SQL Query Rewriting

Problem: Production SQL workloads contain inefficient queries; manual rewriting is costly and does not scale.

Approach:

- Template-based workload abstraction
- Metadata- & DBMS-aware rewrite generation
- Deterministic verification on downsampled data

Result: Significant speedups on JOB, STATS, Stack-Overflow across PostgreSQL & DuckDB.

Publication

PVLDB 2026